Internship: Nasscom SmartBridge - Salesforce Administrator with AI Agentforce

Ganpat University

Team Lead : Itisha Desai Er. No. : 22162171006

Team Member : Nitish Marker Er. No. : 23162582009

Project Name: AI-Powered Customer Support & Service Automation

Platform: Salesforce & Skillwallet

Final Project Documentation

Note : I’ve made this final documentation which covers all the topic according to guidelines but I request the reviewer of project or superwiser to review the other documentations as well which explains project work in details and contents screenshots phasewise. The other documents are also in same github repo. https://github.com/Itisha05/Nasscom-Project.git

1. Project Introduction & Objectives

The AI-Powered Customer Support & Service Automation project is designed to modernize inspection and support workflows by replacing manual, decentralized data entry with an intelligent Salesforce-based solution. The primary goal is to leverage generative AI and low-code automation to reduce administrative overhead while ensuring high data integrity and faster response times.

Key Objectives:

- Centralized Platform: Transition from manual documentation to a secure, cloud-based Salesforce repository.
- AI-Driven Efficiency: Utilize Agentforce and Prompt Builder to automate the generation of structured inspection summaries.
- Automated Lifecycle Management: Implement Record-Triggered Flows to route cases dynamically to specialized support teams.

2. Requirement Analysis & Planning

A comprehensive analysis was conducted to map business needs to technical Salesforce features.

2.1 Functional Scope

- Record Capture: Ability for users to record inspection dates, status, and specific observational notes.
- AI Triggering: Integration of "Field Generation" templates that inspectors can trigger directly from a record page.
- Security & Governance: Implementation of role-based access control and validation rules to protect sensitive data.

2.2 Stakeholder Mapping

- Inspection Officers: Primary users who record data and rely on AI summaries for reporting.
- Salesforce Administrators: Responsible for environment configuration, object schema, and AI template maintenance.
- Management Team: Utilize AI-generated insights and dashboards for operational decision-making.

3. Backend Development & Data Model

The project architecture is built upon a custom schema designed to ground AI responses in accurate business data.

3.1 Custom Object Schema

The following objects were constructed to facilitate the support workflow:

- Customer (Customer__c): Stores unique client profiles.
- Merchandise (Merchandise__c): Tracks product details and service eligibility.
- Order (Order__c): Links cases to specific transactions via Lookup Relationships.
- User Case (User_Case__c): The core entity for tracking customer interactions and storing AI output.
- Knowledge (Knowledge__c): Repository for support articles used to ground AI responses.

3.2 Data Integrity (Validation Rules)

To ensure the AI operates on high-quality data, strict validation rules were implemented on the User Case object:

- Damage_Requires_Priority: Forces a priority selection if the issue category is "Damage".
- Warranty_Delivery_Requires_Priority: Ensures "Warranty" or "Delivery" cases are not saved without a priority level.

4. Support Automation & AI Integration

This section details the "intelligence" layer of the system, combining Einstein AI with automated flows.

4.1 AI Agentforce Configuration

- Activation: Einstein Generative AI was enabled to allow the use of Large Language Models (LLMs).
- Prompt Builder: A Field Generation template named MetaDataInspection was created.
- Prompt Logic: The template uses dynamic resources (CreatedDate, Name, Case_Status__c) and the OpenAI GPT 4o Omni Mini model to generate professional summaries.
- AI Write-Back: Generated content is automatically saved to the AI_Summary__c field on the User Case record.

4.2 Flow Automation

A Record-Triggered Flow serves as the system's engine:

- Assignment: Automatically sets the initial status to "Open".
- Decision Logic: Evaluates the case type (Damage vs. Warranty) for specialized routing.
- Queue Routing: Dynamically retrieves Queue IDs to route records to the Damage Support or Warranty Support groups.
- SLA Check: Includes an "Is High Priority" decision to accelerate critical cases.

5. UI/UX Customization

The user interface was branded and optimized for support agent efficiency using the Lightning App Builder.

- App Branding: A custom app named AI-Powered Customer Support & Service Automation was created with a professional hex color #007002.
- Navigation: The bar was customized for one-click access to Customers, Merchandise, Orders, User Case, Reports, and Dashboards.
- Visual Identification: Unique icons (People, Box, Circle, Chip, Scales) were assigned to tabs for clear UI navigation.

6. Maintenance & Deployment Strategy

Post-deployment stability is ensured through a proactive maintenance framework.

- Monitoring: Continuous tracking of case assignment accuracy and AI response effectiveness through system logs and reports.
- Troubleshooting: Diagnosing failed automations or permission errors, with fixes applied in sandbox environments before production release.
- Deployment: Final configurations were activated, and user training was provided for record creation and AI summary generation.

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7. Conclusion

The AI-Powered Customer Support & Service Automation project successfully demonstrates the synergy between Salesforce administration and Agentforce AI. By integrating custom object relationships with automated flows and generative AI, the system provides a scalable, secure, and intelligent solution that significantly improves operational response times and data continuity.

The **AI-Powered Customer Support & Service Automation** project successfully demonstrates how Salesforce can be used to automate case management and enhance customer service efficiency. By implementing validation rules, automation flows, approval processes, security controls, and AI-assisted support, the system reduces manual effort and improves response time.

This project highlights the effective use of **Salesforce** to deliver a scalable, secure, and intelligent customer support solution while providing valuable hands-on experience with real-world business automation.